Machine–Human Realization Signatures from Surface Form to Hidden Response

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1 Introduction

The familiar "AI feel" of generated prose conflates three questions: which observable linguistic choices differ from human writing, where those choices are readable inside a model, and whether moving the corresponding hidden coordinate changes generated language. Prior work documents lexical, syntactic, and rhetorical differences (Zamaraeva et al., 2025; Zanotto and Aroyehun, 2025; Reinhart et al., 2025) and develops probing and activation-steering tools (Hewitt and Manning, 2019; Rimsky et al., 2024). The missing link is an identification chain connecting an explicit surface measurement to a hidden intervention and then testing its realized effect.

We study a narrower object than universal machine authorship: a *machine–human realization signature*, or recurrent lexical and syntactic choices made while continuing the same natural context. Two complementary 500-pair protocols isolate it. The proposition-conditioned protocol gives a generator a natural prefix and a proposition plan extracted from a matched human continuation, improving semantic alignment but introducing an asymmetric plan-to-text interface. The prefix-only protocol retains the sample, prefix, human reference, generator, and target length while withholding the plan and human text. Agreement across these protocols supports a recurring realization contrast rather than an artifact confined to explicit plan verbalization. No detector or human "AI-likeness" rating defines the target.

A concrete pair. One literary sample illustrates the distinction. The human continuation asks, "Why prematurely claim to have reached finality, when unexpected novelties may shatter any system before it is even completed?" The matched machine continuation states, "Claiming finality is premature because unexpected novelties can shatter any system before completion." The proposition is preserved, but a rhetorical question becomes a declarative clause, the relation is lexicalized with *because*, and pronominal and function-word material is reduced. No individual pair defines our construct. Its role is to make the object concrete: we measure recurring choices of realization after controlling topic and context as far as each interface allows, rather than asking whether a detector can separate arbitrary human and machine documents.

Why the chain is difficult. Each transition in this question introduces a different confound. A surface contrast may reflect topic, domain, generator identity, or the interface used to elicit a continuation. A hidden-state predictor may then exploit source separation without encoding the graded linguistic variable of interest. Finally, an intervention can strongly move a fitted readout while leaving the sampled linguistic measurements unchanged, because the fixed hidden trajectory and the autoregressive generation process are different estimands. We address these risks in order: environment and interface transfer for the surface construct, cross-source and cross-generator prediction for the readout, a fitted-pipeline permutation null for hidden transport, and a separately sealed generation experiment for realization. This ordering is what turns three individually familiar tools—linguistic measurement, probing, and activation intervention—into one falsifiable evidence chain.

We then ask whether the surface scores have a reproducible hidden route. A late-layer readout must transfer to unseen generators and predict both human and machine continuations. We translate that readout to an early perturbation with the model Jacobian, construct human- and machine-side pullbacks independently, and retain their normalized consensus. A matched null permutes the surface-score correspondence within corpus–generator strata and repeats the entire readout–pullback–fixed-token pipeline. This is stricter than

comparing a fitted direction with an ordinary random vector, which lacks the same fitting and pullback advantages. The stages deliberately answer noninterchangeable questions. Surface stability does not imply that a model represents the same score; a hidden readout does not establish that the coordinate is causally used; and a fixed-token causal response does not show that autoregressive generation realizes the corresponding linguistic change. We therefore test each transition and preserve negative outcomes instead of defining success by the final generated text.

The resulting lexical trace exceeds every one of 100 complete null pipelines in both Llama 3.1 8B and Qwen3 4B. The syntactic trace does not. Crucially, fixed-token hidden response does not imply reliable free-generation control: lexical selectivity remains unresolved, and a one-seed Qwen syntactic inversion changes sign in an independent three-seed replication. Our contribution is therefore both positive and negative: an automatically measured linguistic signature can have a reproducible causal hidden trace without becoming a portable, component-selective control direction. This contrast is the paper's central result, not a failed steering demonstration: it localizes where an apparently coherent surface-to-behavior story ceases to be supported.

Contributions. We provide (i) a two-interface surface construct audited across generators, domains, semantic alignment, redundancy, and plan-serialization risk; (ii) cross-generator hidden readouts that transfer from proposition-conditioned training to plan-free outputs; (iii) a canonically oriented, source-consistent Jacobian pullback evaluated against a complete fitted-pipeline null rather than an unmatched random vector; and (iv) a trajectory-distinct generated-text audit with a full cross-component response matrix, orthonormal behavior controls, an independent three-seed replication, and content/quality guardrails. Appendix A provides the complete experimental specification and routes each condensed claim to its full table.

2 Related Work and Scope

Machine-generated language differs from human text across vocabulary, dependency structure, and discourse organization, but many proposed cues vary by model and domain (Muñoz-Ortiz et al., 2024; El Attar et al., 2026; Ahmad et al., 2026). Detector transfer failures likewise caution against treating any corpus-specific contrast as an authorship universal (Sadasivan et al., 2024; Li et al., 2024; Shamardina et al., 2025). We therefore require generator and domain stability and never evaluate a detector.

Representation-based detectors also find that human and generated text can occupy separable regions in observer-model hidden states (Chen et al., 2025; Vishnyakov and Gaintseva, 2026). That objective differs from ours in two ways. Classification only requires a source boundary, whereas our readout must predict a continuous, independently measured linguistic score on both source sides. Moreover, a useful detector need not supply an intervention with a linguistically defined positive pole. We therefore treat source separability as a potential shortcut and require human–machine agreement before constructing the consensus readout.

A probe establishes accessibility rather than causal use (Hewitt and Manning, 2019); intervention reliability additionally depends on alignment between a readout and the behavior being changed (Canby et al., 2025; Waldis et al., 2026). Contrastive activation addition and related methods can control diverse behaviors (Turner et al., 2023; Rimsky et al., 2024; Li et al., 2023), but specificity and direction identifiability remain open (Goyal and Daumé III, 2026; Venkatesh and Kurapath, 2026). Our target is independently defined in surface measurements before any hidden direction or generation outcome is observed. Nullspace and amnesic interventions further show that causal conclusions depend on what variation an intervention removes or preserves (Ravfogel et al., 2020; Elazar et al., 2021). Our use of a frozen nuisance nullspace has the narrower purpose of removing dominant within-class variation before the linguistic direction is fitted. The contribution is consequently not a new steering primitive in isolation, but a design that makes a declared surface variable survive successively stronger transfer and intervention tests.

3 Data and Surface Signature

Two matched interfaces. The proposition-conditioned corpus contains 500 English continuation pairs, balanced across PG-19, WikiLingua, Wikipedia, and XL-Sum (Rae et al., 2020; Ladhak et al., 2020; Wikimedia Foundation, 2023; Hasan et al., 2021). A planner records 3–10 atomic propositions, required entities, and an endpoint from

the human continuation. A distinct generator—Qwen3.6-Flash, DeepSeek-V4-Flash, GLM-5.2, or MiniMax-M2.5—sees only the prefix and plan. A third model checks coverage, unsupported claims, and contradictions. Planner, generator, and judge are distinct for every pair. Malformed, meta, refused, excessively copied, and extreme-length candidates are rejected and regenerated rather than edited. The prefix-only corpus regenerates every sample using only its prefix and length target. Independent semantic cosine is 0.925 under proposition conditioning and 0.727 under prefix-only generation, versus 0.469 after within-source derangement. This gap is expected and useful. The first arm supplies the stronger content control for defining the signature; the second removes every human-derived proposition, entity, and endpoint from the machine interface. Claims that survive both are less easily explained by either semantic mismatch or plan serialization alone. Full construction, rejection, and leakage audits appear in Appendix A.1.

Frozen measurements. We predefine 35 measurements: 10 lexical, 12 syntactic, and 13 discourse features, plus seven nuisance variables covering length, entity, and number statistics. Lexical and syntactic axes are confirmatory targets; discourse is retained only as an off-target audit because it fails the same stability gate. The measurement dictionary, all 35 effects, redundancy analysis, and PCA sensitivity are in Appendix B.1.

Projection-based acquisition of lexical evidence

Surface coordinate. Within each training fold, group g is residualized against nuisance statistics $n(x)$, standardized, and projected onto the paired direction:

$$\widetilde{f}_g(x) = \mathrm{zscore}_{\mathrm{tr}}[f_g(x) - \widehat{B}_g n(x)],$$
$$u_g = \mathrm{unit}\,\mathbb{E}_{\mathrm{tr}}[\widetilde{f}_g^M - \widetilde{f}_g^H], \qquad s_g(x) = \widetilde{f}_g(x)^\top u_g.$$

Contextual token field. For model m, layer k, and generator e, a nuisance-nullspace direction $v_{m,k}^{(-e)}$ is fitted without generator e. Every held-out token state is then assigned $q_{m,k,t} = \langle h_{m,k,t}, v_{m,k}^{(-e)} \rangle$, centered within its own continuation and scaled by a pair-balanced document standard deviation.

Same-form test. For an exact tokenizer token w present in both members of at least ten matched pairs, $\Delta(w) = \mathbb{E}[q^M(w) - q^H(w)]$ measures contextual relocation, while $a(w) = \frac{1}{2}\mathbb{E}[q^M(w) + q^H(w)]$ locates w in a source-balanced hidden vocabulary field. Bootstrap intervals, BH-FDR, and generator/domain sign agreement replace a frequency ranking.

Across three models, 2.44–2.48M token states per model yield 612–704 exact same-token contrasts; 89–379 pass the FDR/stability gate. Individual forms in Figure 1 are concordances, not estimator coordinates.

Surface results. Figure 1 shows a distributed contrast rather than one marker. It also makes the informal "AI feel" concrete.

The paired rewrites show the associated rhetorical regularization: "Yet he stood on his feet" becomes "Despite this grievous injury, he pressed on," and "you should consider if the beginning section flows well" becomes "It is crucial to check whether the beginning flows naturally." The largest lexical shifts are lower function-word rate ($d = -1.221$), pronoun rate (-0.770), and content-word frequency (-0.759), with greater lexical diversity ($+0.659$). Syntactic differences include shorter dependencies (-0.654) and lower sentence-length variation (-0.491). Lexical and syntactic generator-holdout cosines are at least .967 and .944; their domain minima are .934 and .546. On prefix-only outputs, their original-axis cosines remain .885 and .869, with positive projections for 88.0% and 79.6% of pairs. The discourse direction is less stable across domains and is not promoted to a confirmatory hidden target. A post-hoc plan-overlap split and a PCA-truncation audit retain the lexical and syntactic conclusions, reducing the possibility that they are driven only by serialized plan phrases or repeated correlated coordinates. All environment folds and the plan-removal transfer appear in Appendix Figure 6; the appendix measurement dictionary points to the complete frozen effect file, and Appendix B.5 gives the projection-qualified vocabulary field and extended sentence concordance.

4 Method

4.1 Surface-to-hidden construction

A downstream readout acts as a ruler: its projection is larger when a hidden state predicts more of the measured machine-oriented surface signature. That late-layer ruler cannot simply be inserted at an earlier layer because the two residual streams use different coordinates. The Jacobian transpose translates the ruler backward into an early perturbation predicted to move its reading. We perform this translation independently on human and machine sequences; their agreement is a prerequisite, not something created by flipping one side. The resulting direction is tested first on unchanged token sequences, where movement of the ruler can be isolated from changes in sampled text.

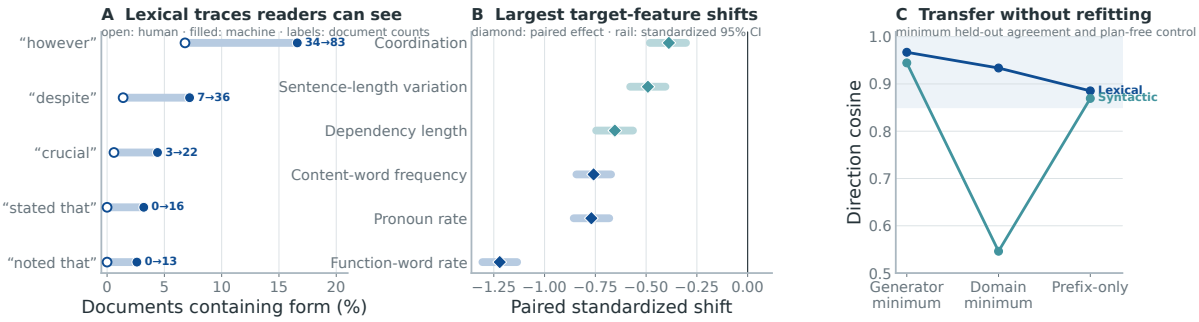


Figure 1: **Measured machine–human realization signature.** (A) Human-to-machine prevalence shifts for concrete forms; labels are machine document counts out of 500. (B) The six largest lexical and syntactic target-feature effects with standardized 95% intervals. (C) Minimum generator and domain holdouts together with agreement after removing the proposition-plan interface. Lexical structure is highly stable, syntax is less uniform across domains, and discourse is not promoted to a causal target.

4.2 Cross-generator readout

Let h_i^k be a continuation-mean hidden state after block k, after projecting away the leading ten pooled within-class nuisance directions. For each held-out generator, source side a, and surface score z_{ga}, we fit the centered forward covariance pattern $\tilde{v}_{ga}^k = H_a^\top z_{ga}/(z_{ga}^\top z_{ga})$. The human and machine patterns must have positive cosine; their normalized sum v_g^k is evaluated by Pearson correlation on both source sides of the unseen generator. All nuisance fitting, standardization, surface directions, and hidden patterns exclude that generator.

The concrete pair through the chain. For the opening literary pair, machine minus human projects $+2.30$ on the frozen lexical surface direction and $+2.18$ on the syntactic direction. Its Llama L32 hidden-state difference nevertheless projects -0.77 on the globally fitted lexical readout. One example can therefore disagree with the aggregate relationship. The pair is used only among 32 balanced Jacobian-fit pairs; the consensus direction is evaluated on 100 different sealed sequences. This separation illustrates why examples explain the objects but folds and sealed trajectories supply the evidence.

4.3 Source-consistent pullback

At early layer ℓ, the prompt-level translation of the frozen late-layer readout is

$$\begin{aligned} a_{ig}^a &= (J_i^{\ell\to k})^\top v_g^k, \\ \hat{a}^a &= \mathrm{unit}(P_m \bar{a}^a), \\ w_{mg} &= \mathrm{unit}(\hat{a}^{\textsc{Human}} + \hat{a}^{\textsc{Machine}}), \end{aligned} \tag{1}$$

where P_m is the frozen nuisance-nullspace projector. We never flip either source-side pullback. Thus $+w$ is canonically predicted to increase the machine-oriented readout rather than being oriented by generated outcomes. On unchanged token sequences, symmetric $+w/-w$ intervention measures

$$\tau_i(w,v) = \frac{(h_i^k(+\alpha w) - h_i^k(-\alpha w))^\top v}{2\alpha\|w\|_2}.$$

Signed τ tests the learned pole; absolute capacity $C_i = |\tau_i|$ permits comparison with null targets that have no semantic orientation.

4.4 Matched null and fixed-token test

Each of $R = 100$ replicates permutes surface scores within corpus–generator strata, refits every layerwise readout, reconstructs its human/machine-consensus pullback, and intervenes on the same 100 hashed token sequences. The confirmatory statistic is depth-normalized area across five intermediate layers; injection and fitted readout-anchor layers are excluded. The add-one rank $p = (1 + \#\{T_r^0 \geq T\})/(R+1)$ treats complete pipelines, not layers or prompts, as null draws.

Frozen protocol. We test Llama 3.1 8B-Instruct and Qwen3 4B at intervention layers L7 and L8, with final sampled readout anchors L32 and L36. Directions use 32 balanced Jacobian pairs; perturbation norm is 5% of the layer-state norm. Free generation uses 166 disjoint prompts, 128 tokens, temperature .7, top-$p = .95$, paired $+1/-1$ seeds, and a separate three-seed replication. Choices are frozen before generation outcomes. Appendix A.2 gives layer conventions, equations, selection firewall, estimands, and all sensitivity settings.

4.5 Generated-language realization

The pullback maximizes local change in a hidden readout along a fixed trajectory. Expected generated-surface change instead depends on the gradient of the sampling distribution and its alignment with that pullback. Autoregressive decoding can amplify, cancel, or reverse a small early change, and a global direction averages heterogeneous prompt-specific routes. We do not estimate this behavior gradient directly; the distinction motivates a trajectory-distinct generation experiment rather than treating fixed-token response as a behavioral result. For each prompt, learned and comparison directions share decoding settings, paired random seeds, and the same $+1/-1$ intervention magnitude. The target estimand is the paired change on the direction's own frozen surface score; component selectivity subtracts the mean change on the other two measurement groups. Prompts, direction poles, strengths, layer sites, and all reporting rules are fixed before outcomes are inspected. The one-seed response matrix and eight orthonormal behavior controls are followed by an independent three-seed learned-axis replication; seed rows are reported as decoding diagnostics, while aggregate intervals resample prompt means after within-prompt seed averaging.

5 Results

Readability and transfer. All 16 held-out-generator readout observations are positive (Table 1A). Mean proposition-conditioned correlations range from .395 to .450; the same readouts, fitted without prefix-only outputs, obtain means of .421–.544 on plan-free machine text. Human/machine pullback cosines are .916–.942, whereas pullback/encoding cosines are only .193–.265. The direction that reads a property and the earlier perturbation that reaches that readout are therefore distinct objects.

A lexical but not syntactic matched trace. Llama lexical AUC is .368 versus a null mean of .201; Qwen is .315 versus .214. No complete null pipeline reaches either learned curve, giving $p = 1/101 = .0099$. The signed lexical response is machineward on all 1,000 model–layer–sequence cases, so the absolute test does not conceal an arbitrary learned pole. Prompt-bootstrap learned-minus-null intervals are positive at every intermediate lexical layer. At the fitted final anchor, however, both learned and null responses can be large; this is why the anchor is excluded from the confirmatory curve. Syntactic AUCs are .255 and .253, with 10 and 19 null curves at least as large ($p = .1089, .1980$). The construction can therefore yield a readable and locally responsive syntactic coordinate without exceptional propagation under the fitted null. Syntax is positive on 999 of 1,000 signed responses, yet one intermediate layer in each model favors the null mean. Sign consistency establishes the declared pole; it does not substitute for target-specificity under a complete refitted null. Layerwise estimates, signed counts, comparator trajectories, and estimator sweeps are consolidated in Appendix Figure 4.

Direction and estimator controls. The canonical pullback exceeds the upstream encoding, raw source difference, nullspace source difference, and an eight-direction projected random ensemble in every model–axis case; all paired intervals exclude zero. Rank and fit-count sweeps retain high cosine with the frozen axis. Appendix Figure 4 exposes the comparator AUCs and the complete rank and fit-count stress ranges.

Hidden response does not become selective control. For generated text, target projection is $p_{ig} = s_g(Y_i^{+w}) - s_g(Y_i^{-w})$, and selectivity subtracts the mean response of the two off-target groups. In the sealed one-seed matrix, lexical target intervals include zero in both models. Qwen syntax moves opposite its canonical pole, but its three replication seed means are $-.245, -.064, +.384$; the prompt-clustered mean becomes $.025[-.189, .295]$. The one-seed inversion is therefore a genuine signed observation under the frozen pole, but not a decoding-stable effect. Llama syntax has a positive replicated target projection $.249[.043, .518]$, yet selectivity remains unresolved $.152[-.092, .445]$. All four selectivity intervals include zero. Seven of eight target-minus-individual-off-target intervals also include zero. The exception, Llama syntax versus discourse, does not survive comparison with the lexical off-target and therefore cannot rescue a component-selective claim. Semantic similarity remains .864–.875, and most recall, length, fragmentation, and repetition guardrails include zero, ruling out gross collapse without ruling out smaller semantic drift. Appendix Figure 5 reports learned-versus-random calibration and every pairwise selectivity interval; the main figure already shows the complete response profiles and seed means, avoiding a duplicated appendix panel. The accompany-

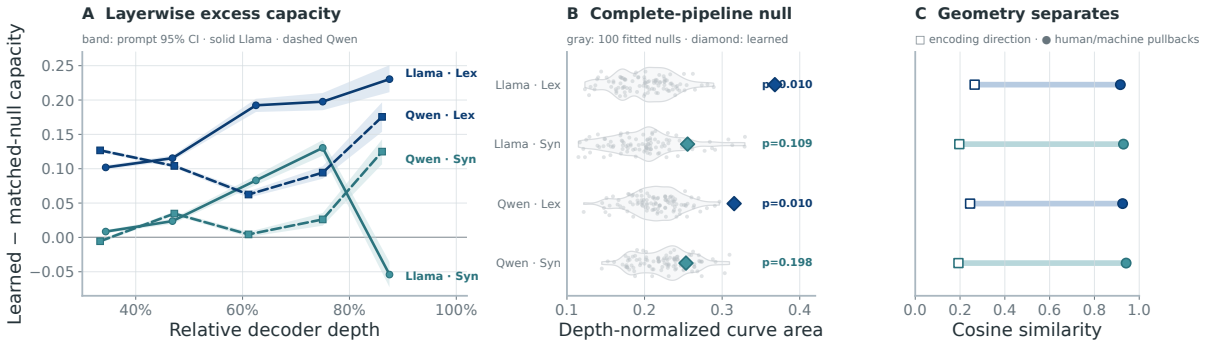


Figure 2: **Surface-to-hidden intervention trace.** (A) Learned-minus-null absolute capacity at five intermediate checkpoints. (B) Complete depth-normalized curve statistics from 100 fitted null pipelines; diamonds are learned curves. (C) Pullback–encoding and human–machine pullback cosines. Injection and readout-anchor layers are excluded.

A. Every held-out-generator fold

Model	L	Axis	Held-out gen.	Human r	Machine r	Mean r	H/M cos.	Fold/full cos.	$\|\Delta u\|_2$
Llama 3.1 8B-Instruct	7	Lex.	MiniMax	.346	.464	.405	.519	.855	1.473
			DeepSeek	.530	.401	.466	.535	.895	1.435
			GLM	.545	.526	.536	.524	.846	1.513
			Qwen	.399	.392	.395	.486	.882	1.213
		Syn.	MiniMax	.381	.360	.371	.607	.852	1.590
			DeepSeek	.337	.446	.392	.433	.910	1.453
			GLM	.356	.396	.376	.576	.786	1.429
			Qwen	.550	.335	.442	.480	.911	1.328
Qwen3 4B	8	Lex.	MiniMax	.355	.388	.372	.325	.939	1.473
			DeepSeek	.459	.423	.441	.487	.941	1.435
			GLM	.459	.592	.525	.468	.936	1.513
			Qwen	.432	.426	.429	.468	.925	1.213
		Syn.	MiniMax	.632	.340	.486	.553	.883	1.590
			DeepSeek	.445	.520	.483	.442	.910	1.453
			GLM	.496	.374	.435	.513	.914	1.429
			Qwen	.478	.313	.396	.563	.927	1.328

B. Transfer, geometry, and matched-pipeline trace

Model	Axis	Plan μ	Prefix μ	H/M cos.	Enc. cos.	Learned AUC	Null μ	p_{null}
Llama 3.1 8B-Instruct	Lex.	.451	.470	.916	.265	**.368**	.201	**.0099**
	Syn.	.395	.421	.930	.197	.255	.202	.1089
Qwen3 4B	Lex.	.442	.544	.926	.245	**.315**	.214	**.0099**
	Syn.	.450	.451	.942	.193	.253	.225	.1980

Table 1: **Central representation-to-transport evidence.** Panel A reports all held-out-generator folds. Each fold trains on 375 and tests on 125 continuations per source; fold/full cosine compares its direction with the full-data direction, and $\|\Delta u\|_2$ is the held-out surface shift norm. Panel B reports plan-conditioned and prefix-only means, direction geometry, and the five-layer fixed-token test.

ing appendix text reports the content and quality guardrails.

6 Discussion

The results separate three claims that are often collapsed. First, machine–human realization is a distributed, corpus-conditioned linguistic contrast, not one detector cue. Second, the lexical contrast maps to a cross-generator hidden coordinate with a matched-pipeline causal trace in both target models. Third, that coordinate is not automatically a reliable control knob for generated language. A pullback optimizes local movement of a hidden readout on a fixed trajectory; generation also depends on the behavior-gradient magnitude, its alignment with the global direction, and autoregressive divergence.

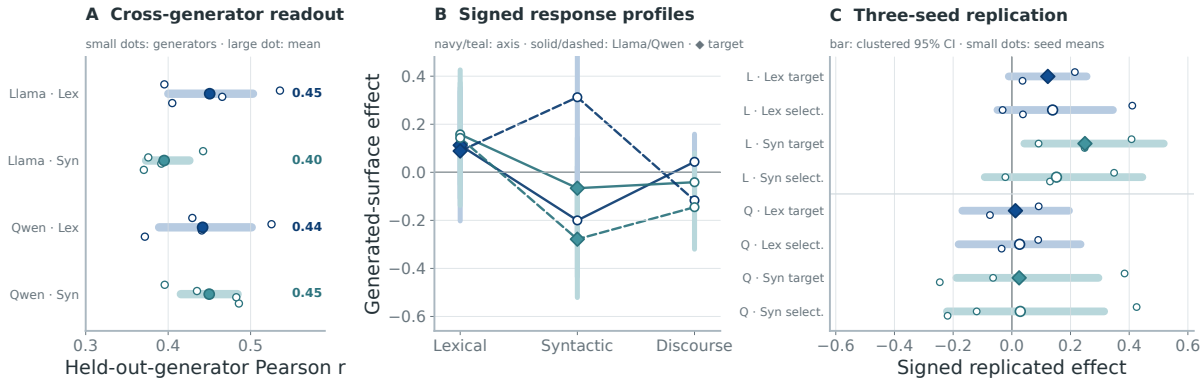


Figure 3: **Representation-to-realization audit.** (A) Cross-generator readouts. (B) One-seed generated-surface response profiles against all measurement groups. (C) Target projection and component selectivity in the independent three-seed replication. Panels B–C are $+1$ minus -1 effects; diamonds mark target components and small points in C are seed means.

Readability, fixed-token transport, and selective realization must therefore be tested separately.

What the evidence chain identifies. The two corpus interfaces play different roles. Proposition conditioning narrows semantic variation and makes a paired realization comparison possible, but it can imprint a plan-verbalization regime. Prefix-only generation removes that regime but no longer guarantees that the machine continuation expresses the same propositions as the human reference. The result that matters is their intersection: lexical and syntactic directions remain aligned, and readouts trained only on proposition-conditioned examples predict plan-free outputs from held-out generators. This does not make every feature content-invariant. It shows that the multivariate directions recur under two protocols with opposite confounds. The discourse inventory fails this standard and is consequently used as an off-target diagnostic rather than silently folded into the target.

Why the matched null changes the claim. An ordinary random input direction is disadvantaged twice: it is neither fitted to a surface target nor translated backward through the model Jacobian. The matched null gives each permuted target both advantages and repeats all source-side agreement and fixed-trajectory steps. The lexical result therefore says more than "this direction moves some hidden activation." It says that the complete five-layer capacity curve is unusual among directions obtained by the same analysis pipeline after breaking only the linguistic correspondence. Conversely, the syntactic result demonstrates the value of this comparator. Syntax is readable, its human and machine pullbacks agree, and its signed responses are predominantly machineward, yet its curve is not exceptional under the fitted null. Those descriptive facts cannot be promoted into a confirmatory propagation claim.

Why transport can fail at realization. The fixed-token intervention estimates how an early perturbation changes a later readout along an unchanged trajectory. Generation replaces that trajectory after the first token whose distribution changes. Even locally, the perturbation that maximizes movement of a hidden readout need not align with the gradient of the expected surface measurement. A single global vector also averages prompt-specific pullbacks whose behavioral effects may differ in magnitude or sign. The three-seed Qwen syntactic result makes this distinction concrete: the canonical pole and fixed-token response are unchanged, while sampling produces two negative seed means and one larger positive mean. The correct conclusion is sampling-sensitive realization, not a post-hoc axis flip. Likewise, Llama's positive syntactic target effect is insufficient because the lexical and discourse measurements move enough that component selectivity remains unresolved.

What the negative result rules out—and what it does not. The semantic and quality guardrails argue against wholesale output collapse as the sole explanation. They do not establish that each paired output preserves all facts, entities, or pragmatic commitments. The negative control result is also construction-specific. It applies to one canonically oriented, single-layer, constant intervention at a frozen strength and decoding regime. Multi-layer

schedules, token-conditional interventions, or directions optimized for generated-surface response could behave differently. Testing those alternatives would answer a new question: whether a behavior-aware controller can realize the signature, rather than whether a surface-readout pullback already supplies such a controller. The present design keeps those hypotheses separate and avoids optimizing on the same generation outcomes used for evaluation.

Implications. For interpretability studies, the practical lesson is to report an evidence ladder rather than one steering score: out-of-environment readability, source-side geometric agreement, matched fixed-token transport, the complete cross-component response matrix, and decode-seed robustness. For linguistic studies, the stable object is a direction over a declared measurement space, not a universal list of machine tells. For detection, these results provide no calibrated authorship boundary: the axes are conditional on corpora, domains, generators, and time, and the paper deliberately does not estimate false-positive rates on a deployment population.

7 Conclusion

We connected a frozen linguistic measurement to hidden representation, fixed-token causal response, and generated language without treating those stages as interchangeable. Lexical and syntactic realization signatures recur across generators and across proposition-conditioned and prefix-only interfaces. Both are readable in two model families, but only the lexical direction produces a hidden transport curve that is exceptional under the complete fitted-pipeline null. Even that positive hidden result does not become reliable, component-selective generation control. The central finding is therefore a boundary: an interpretable surface regularity can have a reproducible causal hidden trace while failing as a portable behavioral knob.

This conclusion is deliberately scoped to English, two model families, four named generators, one single-layer intervention construction, 100 fixed sequences, and 166 generation prompts. It does not identify a universal authorship factor or establish that readers perceive the measured signature as "AI-like." The measurements must not be used as detectors or punitive thresholds. Larger models, languages, intervention schedules, and behavior-gradient-aware directions remain open tests. Appendix A.5 gives the complete limitations, ethics, and reproducibility record.

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A Experimental Design and Protocol Details

The appendix follows the same evidence chain as the main paper. Appendix A.1 expands corpus construction; Appendix A.2 defines the estimands and intervention protocol; the next two sections visualize direction and decoding robustness; and Appendix A.5 records scope, ethics, and reproducibility qualifications. Numerical panels are generated directly from the frozen observation files.

A.1 Corpus Construction and Measurement Details

Proposition-conditioned arm. Human continuations are balanced across PG-19 narrative, WikiLingua instructional text, Wikipedia encyclopedic text, and XL-Sum news, with 125 pairs from each source. A planner sees the natural prefix and human continuation and records 3–10 atomic propositions, required entities, and an intended endpoint. A distinct generator sees the prefix and plan but never the human continuation. Planner, generator, and judge are distinct for every pair. Acceptance requires complete or mostly complete core-content coverage, no major unsupported claim, and no major contradiction. Structural gates reject malformed, list-like, meta, refused, excessively copied, or extreme-length candidates; rejected attempts are regenerated, not edited. These controls establish automatic proposition conditioning, not guaranteed semantic or rhetorical equivalence.

Prefix-only arm. For each accepted sample, the control retains its identifier, source, natural prefix, human continuation, assigned generator, and target word count. The generator receives only the prefix and scalar length request—never the plan, entity list, endpoint, human continuation, or a transformation of them. It passes the same structural gates but no content-overlap gate. This removes the asymmetric planning interface at the cost of weaker semantic matching. Axes and readouts evaluated on this arm are fitted exclusively on the proposition-conditioned corpus.

Measurement and fitting. The frozen inventory contains 10 lexical, 12 syntactic, and 13 discourse measurements. Seven nuisance variables—word, sentence, paragraph, and character counts, mean sentence length, named-entity rate, and number rate—are removed by training-fold regression. Feature standardization, nuisance fitting, surface-axis construction, and hidden readout fitting all occur inside the relevant training fold. The discourse group remains an off-target audit rather than a stable hidden target. Appendix B.1 gives the dictionary, strongest effects, complete environment holdouts, plan-removal control, and PCA sensitivity.

A.2 Complete Estimands and Intervention Protocol

Table 2 separates the evaluation units used at each stage. Construction inputs are never counted as independent test observations: generator folds support transfer, fixed sequences support transport estimation, complete refitted pipelines support the null, and prompt means support generated-language intervals.

Readout. For source side a, let H_a be centered, nuisance-residualized hidden states in the frozen within-class nullspace and z_{ga} the corresponding centered surface scores. The source-side and consensus activation patterns are

$$\begin{aligned} \widetilde{v}^k_{ga} &= H_a^\top z_{ga}/(z_{ga}^\top z_{ga}), \\ v^k_{ga} &= \mathrm{unit}(\widetilde{v}^k_{ga}), \\ v^k_g &= \mathrm{unit}(v^k_{g,h} + v^k_{g,m}). \end{aligned} \tag{2}$$

This is a forward covariance activation pattern, not a high-dimensional backward regression. Human and machine patterns must agree positively before their normalized sum. The reported fold statistic averages the two held-out Pearson correlations; Table 1 retains both source-side values.

Pullback and signed response. For layer-to-layer Jacobian $J_i^{\ell\to k}$, source-side pullbacks are averaged and then projected by the frozen nuisance projector P_m:

$$\begin{aligned} a^a_{ig} &= (J_i^{\ell\to k})^\top v^k_g, \\ \hat{a}^a &= \frac{P_m \bar{a}^a}{\|P_m \bar{a}^a\|_2}, \\ w_{mg} &= \frac{\hat{a}^{\textsc{Human}} + \hat{a}^{\textsc{Machine}}}{\|\hat{a}^{\textsc{Human}} + \hat{a}^{\textsc{Machine}}\|_2}. \end{aligned} \tag{3}$$

Neither source-side vector is sign-flipped. The declared $+w$ pole is inherited from the machine-oriented readout. For a fixed token sequence,

$$\tau_{img}(w) = \frac{(h_i^k(+\alpha w) - h_i^k(-\alpha w))^\top v_g^k}{2\alpha \|w\|_2}. \tag{4}$$

Signed τ gives directionality; $|\tau|$ gives capacity comparable with unoriented null targets.

Stage	Estimand	Construction and evaluation
Surface	Axis u_g	500 paired feature differences; Δ, paired d, and FDR q.
Readout	Cross-generator D	Three-generator train, one-generator test; Pearson r on both sources.
Pullback	Canonical w	32 balanced Jacobian pairs; source-side agreement and frozen-anchor response.
Trace	Capacity C	Five intermediate checkpoints on 100 identical fixed sequences.
Null test	Curve p_{null}	100 complete pipeline refits and an add-one curve-statistic rank.
Generation	Target p_g	Paired $+1/-1$ generations; prompt-bootstrap target effect over 166 prompts.
	Selectivity b_g	Target response minus the mean of two off-target responses.

Table 2: **Estimands and evaluation units.** The leftmost column contains the largest row grouping, so its multirow label is both vertically centered and visually primary. Held-out generators, fixed sequences, complete null pipelines, and prompt-level paired effects answer different questions and are not pooled into one sample size.

Matched pipeline null. Each replicate permutes surface scores within corpus–generator strata, refits every layerwise readout, refreezes the global direction, repeats the Jacobian pullback, and evaluates the identical hashed trajectories. Writing $C_i(w, v) = |\tau_i(w, v)|$,

$$\widehat{C}_{mgj} = \frac{1}{n}\sum_i C_i(w_{mg}, v_g^j),$$
$$\widehat{C}^0_{mgj} = \frac{1}{R}\sum_r \frac{1}{n}\sum_i C_i(w_{mgr}, v_{gr}^j). \quad (5)$$

We use $n = 100$, $R = 100$, five intermediate layers, and a depth-normalized curve statistic. Injection and fitted readout-anchor layers remain descriptive construction-internal diagnostics.

Generation and selection firewall. Hooks begin after prompt prefill. Fixed-token tests preserve the sequence across learned and null directions; free generations may diverge autoregressively. Intervention layer, endpoint, norm, pole, prompts, and decoding parameters are frozen before outcomes. The sealed response matrix and eight-axis calibration use one paired seed; the independent learned-axis replication uses three further paired seeds. For response $p_{ic}(w)$ on measurement group c, selectivity is

$$b_{ig}(w) = p_{ig}(w) - \frac{1}{2}\sum_{c\neq g} p_{ic}(w). \quad (6)$$

Three-seed intervals average paired effects within prompt before resampling the 166 prompt means.

A.3 Hidden-State Construction and Robustness

Figure 4 replaces the former comparator and estimator tables. Its three panels keep the four model–axis cases aligned, so construction alternatives and every nuisance-rank or fit-count setting can be compared without scanning repeated numeric rows.

The canonical pullback exceeds upstream encoding, raw source contrast, nullspace source contrast, and the projected-random ensemble in all four model–axis cases. The smaller gap to upstream encoding is substantively different from the large gaps to the other controls: it supports the source-consistent pullback geometry, but does not itself establish generated-language control. All nuisance-rank and fit-count variants remain close to the frozen direction. The complete fixed-token trajectory and matched-null distribution remain in Figure 2; repeating them here would add a second rendering of the same observations rather than another robustness test.

A.4 Generated-Language Robustness

Figure 5 complements the response and replication profiles already shown in Figure 3: it asks whether the learned magnitude is exceptional among eight controls and whether the target effect is larger than each individual off-target response.

The seed view makes the negative conclusion visible rather than burying it in a large table. Qwen syntax changes from two negative seed means to a larger positive third seed; its aggregate target and selectivity intervals include zero. Llama syntax is the only aggregate target interval above zero, but its selectivity interval also crosses zero. Lexical effects remain unresolved in both models. Seven of eight pairwise selectivity intervals include zero. Seeds therefore expose decoding heterogeneity and are not counted as independent experimental replications.

A.5 Limitations, Ethics, and Reproducibility

Limitations. The construct is English, automatic, and tested in two target-model families. It cannot attribute differences to architecture, scale, or training, nor establish persistence in larger or closed sys-

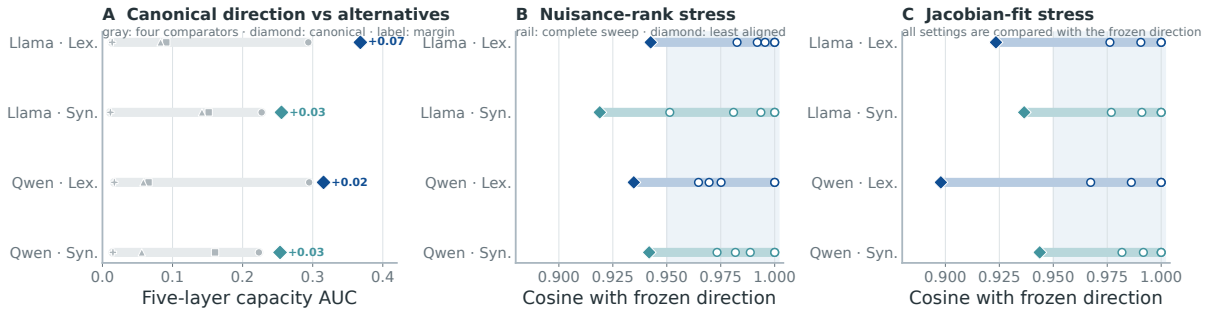


Figure 4: **Hidden-direction construction stress tests.** (A) Canonical five-layer capacity AUC against upstream encoding, raw source contrast, nullspace contrast, and projected-random alternatives; labels give the margin above the strongest comparator. (B–C) Complete cosine ranges under nuisance-rank and Jacobian-fit-size sweeps; diamonds mark the least-aligned setting in each sweep. $k = 10$ and 32 balanced pairs define the canonical setting.

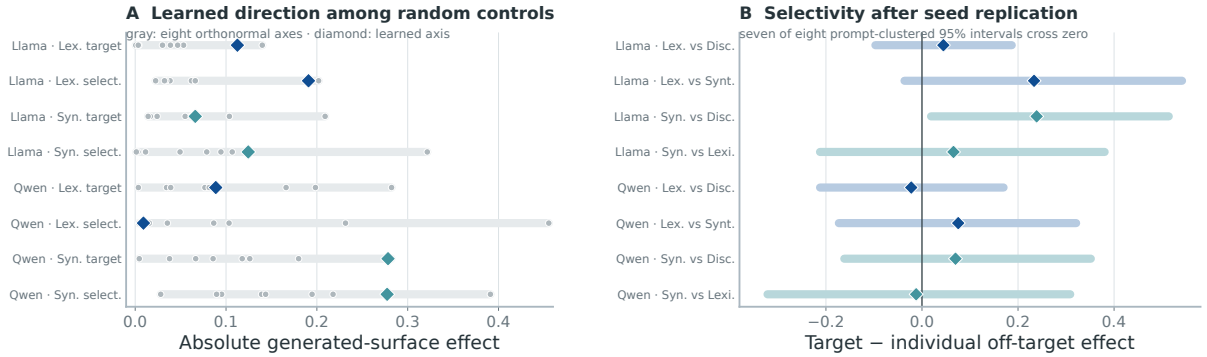


Figure 5: **Generated-language calibration and selectivity.** (A) Absolute learned-axis target and selectivity effects against eight orthonormal controls. (B) Target-minus-individual-off-target effects after the independent three-seed replication. Diamonds are learned estimates; pale rails in B are prompt-clustered 95% intervals and the vertical line marks zero.

tems. Proposition conditioning improves semantic matching but introduces an asymmetric plan-to-text interface; prefix-only generation removes that interface but weakens proposition control. Neither identifies a content-invariant authorship factor. Four generator environments support replication across those environments, not precise population inference. Dependency features inherit parser errors, and shallow discourse markers may understate rhetorical organization. No perception study establishes that the measured forms cause an "AI-like" impression.

Layerwise traces sample checkpoints rather than every decoder block, and the 32 pullback-fit pairs are constrained by Jacobian cost. Rank and fit-count sweeps establish geometric stability but do not replace an independent full refit on a second trajectory set. The one-seed matrix, three-seed replication, fixed strength, and one decoding regime do not characterize the full autoregressive distribution. Intervals containing zero do not prove exact behavioral nulls. The negative behavioral conclusion applies to the declared canonical single-layer pullback, not every possible steering method.

Ethics. The study uses public corpora and automatic continuations; it contains no private conversation or human-subject annotation. The signatures are conditional corpus contrasts and must not be used as authorship detectors. Treating them as norms could unfairly flag non-native, neurodivergent, formal, or technical writing. The release contains no trained detector, authorship threshold, or punitive decision rule. Any operational use requires contextual human review and a meaningful route for affected writers to contest decisions.

Reproducibility. The artifact records dataset revisions and hashes, accepted and rejected attempts, prompt indices, model/layer conventions, direction hashes, random seeds, strengths, and per-prompt observations. English measurements use spaCy 3.8.7 with `en_core_web_sm` 3.8.0. Every fold fits nuisance regression, standardization, surface directions, and hidden readouts on training data only.

Matched nulls preserve source, domain, generator, hidden states, fitting steps, and fixed-token trajectories while breaking target correspondence. Primary statistics and figures regenerate from frozen observations. Recorded confirmatory stages total 3.345 GPU-process hours; provider-side accelerator time for hosted corpus-generation APIs is unavailable. Table 3 gives model-level time, tokens, and work units.

B Supplementary Evidence

This section retains the measurements needed to audit the claims without repeating every frozen value as a typeset row. Effects and robustness checks that are read as comparisons are shown as figures; tables are reserved for the feature definition and compute ledger, where exact field lookup is the primary task.

B.1 Surface Evidence and Concrete Realizations

What "AI-like" means in these pairs. The contrast is visible in recurrent choices, not only in an abstract projection. Across 500 matched pairs, *however* occurs in 83 machine documents and 34 human documents; *these* in 116 and 69; *during* in 78 and 49; *despite* in 36 and 7; and *following* in 35 and 14. Evaluative forms also separate the sources: *crucial* and *significant* each occur in 22 machine documents but only 3 human documents. The reporting frames *stated that* and *noted that* likewise occur in 16 and 13 machine documents and in no matched human document. These are transparent corpus descriptions, not necessary or sufficient conditions for machine authorship.

Paired sentence-level evidence. The same tendency appears as rhetorical repackaging within proposition- conditioned pairs. A human news continuation begins, "In a statement, RBS chief executive Ross McEwen said . . . ," whereas its machine mate writes, "RBS chief executive Ross McEwen stated that . . . ." A human narrative coordinates survival with "Yet he stood on his feet," while the machine makes the relation explicit: "Despite this grievous injury, he pressed on." An instructional sentence changes from "You should consider if the beginning section flows well" to "It is crucial to check whether the beginning flows naturally." The examples instantiate the measured shifts toward explicit transitions, nominal or evaluative framing, and more regular clause packaging. They are selected to interpret frozen measurements, not used to fit them.

Figure 6 makes two asymmetries immediate. The lexical contrast is both larger and more distributed than the discourse contrast, while syntax is characterized mainly by shorter dependencies, less sentence-length variation, and less coordination. The complete 35-row effect inventory is not redrawn as a low-information figure: exact raw differences, standardized effects, FDR-adjusted q-values, and feature names remain in `feature_statistics.csv`.

Lexical and syntactic directions remain stable under generator holdout. Their domain minima are lower—especially the .546 syntactic cosine in the encyclopedic holdout—but remain positive. The discourse direction collapses in its weakest environments and under aggressive PCA truncation, which is why it is retained as an off-target measurement rather than promoted to a hidden causal target. Prefix-only transfer answers a different concern: the lexical and syntactic readouts remain predictive when the generator never receives the human-derived proposition plan.

B.2 Measurement Dictionary and Corpus Audit

Frozen dictionary. The 10 lexical coordinates are function-word, pronoun, contraction, repeated-trigram, abstract, evaluative, emphasis, and institutional-lexicon rates, MATTR-50, and mean content-word Zipf frequency. The 12 syntactic coordinates are mean dependency depth and length; clause, coordination, and subordination densities; passive-clause, nominalization, fragment, apposition, and parenthetical rates; sentence-length coefficient of variation; and adjacent POS parallelism. The 13 discourse coordinates cover six connective families, explicit relations, transition openings, enumeration and summary markers, entity and subject continuity, and adjacent content overlap. Lexical and syntactic groups are hidden-state targets; discourse is an off-target audit. Every training fold residualizes word, sentence, paragraph, and character counts, mean sentence length, named-entity rate, and number rate.

Corpus ledger. The proposition-conditioned corpus contains 500 accepted pairs, balanced at 125 per domain and generator. All 500 pass the independent automatic judge and maintain distinct planner, generator, and judge models. Fifty genera-

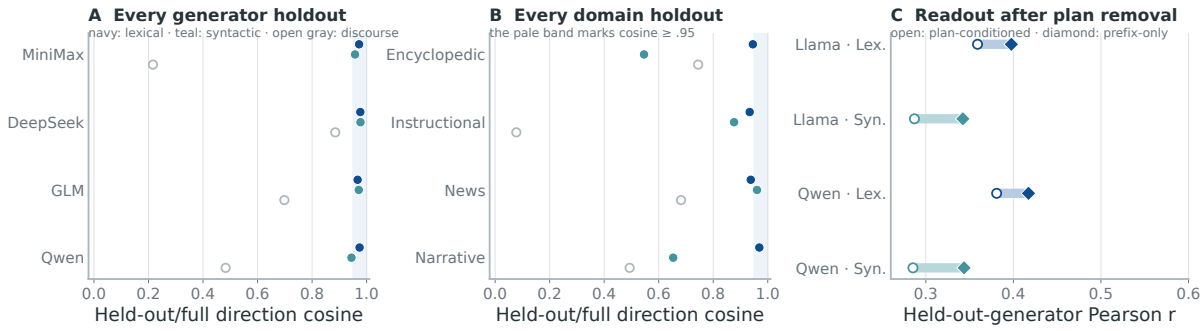


Figure 6: **Surface-axis transfer audit.** (A–B) Direction cosines for every held-out generator and domain. Navy and teal points are the lexical and syntactic targets; open gray points retain discourse as an off-target stability check. (C) Mean cross-generator readout correlations on proposition-conditioned and prefix-only machine text. Open circles are the fitted-corpus values and diamonds are prefix-only values; the prefix-only corpus is excluded from fitting.

tion attempts yield 43 rejected candidate records; accepted texts receive no post-generation repairs. The non-gating plan-serialization diagnostic flags 179 pairs, but the extreme serialization gate flags none. Mean and median machine–human length ratios are .935 and .925, with range .603–1.405. The plan-conditioned semantic cosine is .925, the prefix-only cosine is .727, and within-source derangement gives .469. These records support controlled comparison but do not certify perfect propositional equivalence.

B.3 Generated-Language Guardrails

Figure 5 contains the random-axis and pairwise selectivity comparisons; Figure 3 retains the full target, off-target, and seed views. The paired $+1/-1$ generations remain semantically close: mean paired cosine ranges from .864 to .875 across the four model–axis conditions. Most intervals for content recall, content precision, word count, repetition, and fragmentation include zero. The two clearest nonzero guardrail shifts are modest: Llama lexical steering reduces number recall by .038, and Llama syntax reduces entity recall by .057. These checks rule out gross degeneration, not smaller semantic drift. They also do not rescue selectivity: all four aggregate component- selectivity intervals include zero, and seven of eight target-versus- individual-off-target intervals include zero.

B.4 Compute and Artifact Audit

The compute ledger keeps the model grouping in the leftmost column because it spans the most rows. Light gray partial rules separate stages without cutting through that multirow label.

Model	Stage	GPU-process h	Tokens
Llama 3.1 8B-Instruct	Pullback	0.006	–
	Matched null	0.416	–
	Response + controls	0.857	424,960
	Three-seed replication	0.296	254,976
Qwen3 4B	Pullback	0.004	–
	Matched null	0.313	–
	Response + controls	1.116	424,960
	Three-seed replication	0.338	254,976

Table 3: **Recorded confirmatory compute.** GPU-process hours sum the elapsed time of one model-owning CUDA process. The two target models ran concurrently, so wall time is lower than the 3.345-hour aggregate. Pullback and null stages use 32 paired Jacobians and 100 pipelines $\times$ 100 sequences per concept; response and replication use 10 and 2 direction files. Hosted corpus-generation APIs do not expose provider-side accelerator time; the server snapshot records local hardware and software versions.

B.5 Extended Lexical and Sentence Concordance

Connective and evaluative inventory. The more conspicuous connective choices sharpen the corpus-level description: *consequently* occurs in 21 machine documents and 1 human document, *furthermore* in 11 and 1, *additionally* in 15 and 5, and *ultimately* in 11 and 1. Evaluative closure shows the same asymmetry: *essential* occurs in 16 machine documents and 2 human documents, and *profound* in 7 and 0. The full frame *it is crucial* occurs in 10 machine continuations and no human continuation.

What the inventory does not support. Popular shibboleths are not automatically useful in a matched corpus: *delve* occurs in neither source, *tapestry* occurs once in each, and *landscape* occurs in 6 machine versus 5 human documents. The recognizable contrast here is therefore not a generic blacklist of "AI words." It is the repeated combination of explicit linking, evaluative closure, and regularized clause packaging.

Additional paired excerpts. The four readings below isolate local changes that a list of effect sizes cannot show. Instructions move from personal advice to impersonal prescription; news replaces repeated *said* with differentiated reporting predicates; encyclopedic prose adds explicit stage and evaluation markers; and literary rhetorical questions become declarative causal relations. The pattern is regularization, not a single token shared by every pair.

Projection-qualified vocabulary field. We also project 2.44–2.48M continuation-token states per model onto leave-one-generator-out nuisance-nullspace directions. An exact token is compared only in matched pairs containing it on both sides, after subtracting each continuation's mean coordinate; source separation and frequency therefore cannot create a local shift. Of 612–704 eligible contrasts per model, 89–379 survive BH-FDR and generator/domain sign-agreement gates. The field is model- and tokenizer-local, not a universal authorship lexicon or direct-logit-lens decoding.

Instructional: personal advice → impersonal prescription.

HUMAN: "Consider getting it professionally cleaned rather than doing it yourself."
MACHINE: "Due to their delicate nature, professionally cleaned viscose rugs are strongly recommended."

The recommendation acquires an abstract rationale, passive packaging, and stronger evaluative force.

News: repeated attribution → reporting taxonomy.

HUMAN: "McKeown said . . . 'A major incident is . . . ,' she added."
MACHINE: "McKeown explained . . . she argued . . . She defined a major incident . . . "

The machine assigns a distinct discourse function to each proposition, then links the closing events with *Following* and *Consequently*.

Encyclopedic: event sequence → staged summary.

HUMAN: "After cutting short his stay . . . He helped Penang FA earn second place."
MACHINE: "Subsequently . . . His contributions were instrumental . . . This strong performance secured their promotion."

During his time, *Overall*, and *The following year* further divide the chronology into explicit stages.

Literary: rhetorical question → explicit causality.

HUMAN: "Why prematurely claim to have reached finality?"
MACHINE: "Claiming finality is premature because unexpected novelties can shatter any system."

The proposition is retained, but interrogative stance becomes a declarative claim and the relation is lexicalized with *because*.

Reading note. All examples come from the frozen proposition-conditioned pairs. They interpret the measured inventory; they are not additional fitted features, necessary cues, or authorship rules.